

Lab 3.2: Render a Product Catalogue from Static Data

|
--- | ---

****Goal**** | Turn an array of objects into a grid of components with keys and conditions.

****Suggested time**** | 40 min

This lab practises lists and conditionals on a different domain, so you can see the pattern is universal. You may build it inside TaskBoard in a temporary `src/components/catalogue` folder, or in a second Vite project. The temporary folder is quicker.

Steps

1. In TaskBoard, create `src/components/catalogue/products.js` exporting an array of 8 products, each with an id, name, price, category, inStock and rating. You can use your own, or paste this in to get straight to the rendering:

```
js
export const products = [
  { id: 1, name: "Wireless Mouse", price: 349, category: "electronics", inStock: true, rating: 4 },
  { id: 2, name: "Mechanical Keyboard", price: 1299, category: "electronics", inStock: true, rating: 5 },
  { id: 3, name: "USB-C Hub", price: 599, category: "electronics", inStock: false, rating: 3 },
  { id: 4, name: "Learning React", price: 489, category: "books", inStock: true, rating: 5 },
  { id: 5, name: "JavaScript Pocket Guide", price: 199, category: "books", inStock: true, rating: 0 },
  { id: 6, name: "Desk Lamp", price: 425, category: "office", inStock: false, rating: 4 },
  { id: 7, name: "Notebook Set", price: 89, category: "office", inStock: true, rating: 3 },
  { id: 8, name: "Laptop Stand", price: 749, category: "office", inStock: true, rating: 4 },
];
```

Product 5's rating: 0 is deliberate — keep it, because it surfaces a common bug when you render the rating in step 2 below.

2. Create `ProductCard.jsx` showing the name, price and rating.

3. Show an Out of stock badge only when `inStock` is false.

4. Create `ProductGrid.jsx` that maps products to `ProductCards` with keys, and render it in App for now.

5. Show a No products message when the array is empty. Test it by temporarily passing an empty array (`products=[]`), then restore the real data.

6. Style the grid with CSS Grid.

7. Commit your work. When you move on to Lab 3.3, remove `ProductGrid` from App.

Done when

- 8 cards render in a grid
- Out of stock badges appear on the right products
- An empty array shows the message

Hints (try on your own first)

Format a price as South African rand:

```
new Intl.NumberFormat("en-ZA", { style: "currency", currency: "ZAR" }).format(price)
```

Show a whole-number rating as stars:

```
"★".repeat(rating)
```

Troubleshooting

Problem | Fix

--- | ---

Key warning | The ``key`` is inside ``ProductCard`` (on the element it returns) rather than on the element returned by ``map``: `<ProductCard key={product.id} ... />``.

Stray ``0`` on screen | ``{product.rating && ...}`` when a rating is 0. Compare explicitly: ``product.rating > 0 &&``.

Grid shows one column | ``display: grid`` is missing, or it is on the wrong element (it belongs on the container around the cards).

> Stretch challenge

> - Add a category heading above each group of products by filtering per category.

> - Sort by price with `[...products].sort((a, b) => a.price - b.price)`. Copy with spread first, because sort changes the array in place.